

# ガウス記号のおぼえがき

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## Notation.

$a, b \in \mathbb{N}$ ,  $c, d \in \mathbb{Z}$  とし, 関数  $f: \mathbb{Z} \rightarrow \mathbb{Z}$ ,  $g: \mathbb{Z} \rightarrow \mathbb{Z}$ ,  $h: \mathbb{Z} \rightarrow \mathbb{Z}$ ,  $u: \mathbb{Z} \rightarrow \mathbb{Z}$  を

$$\forall x \in \mathbb{Z}; f(x) := \left[ \frac{ax + c}{b} \right]$$

$$\forall y \in \mathbb{Z}; g(y) := \left[ \frac{by + d}{a} \right]$$

$$\forall x \in \mathbb{Z}; h(x) := g(f(x))$$

$$\forall x \in \mathbb{Z}; u(x) := ax - bf(x)$$

と定める. 更に,  $p, q \in \mathbb{R}$  に対し,

$$J_{p,q} := \{n \in \mathbb{Z} : p \leq n \leq q\}$$

と置く.

## Theorem.

$\forall x \in \mathbb{N}; h(x) = x$  となる為の必要充分条件は

$$\max_{z \in J_{1,b}} u(z) \leq d < a + \min_{z \in J_{1,b}} u(z)$$

である.

## Outline of Proof.

( $\Rightarrow$ )

- $r \in J_{1,b}$  を任意に取る.
- $h(r) = r$
- $r \leq \frac{bf(r) + d}{a} < r + 1$
- $u(r) \leq d < a + u(r)$
- $\max_{z \in J_{1,b}} u(z) \leq d < a + \min_{z \in J_{1,b}} u(z)$

( $\Leftarrow$ )

- $x \in \mathbb{N}$  を任意に取る.
- 除法の原理から,  $\exists! k \in \mathbb{N} \cup \{0\}; \exists! r \in J_{1,b}; x = bk + r$  が従う.
- $h(x) = x + h(r) - r$
- $\max_{z \in J_{1,b}} u(z) \leq d < a + \min_{z \in J_{1,b}} u(z)$

- $u(r) \leq d < a + u(r)$
- $r \leq \frac{bf(r) + d}{a} < r + 1$
- $h(r) = r$
- $h(x) = x + h(r) - r = x$

□

**Lemma.1**

$$\forall x \in \mathbb{R}; \forall n \in \mathbb{Z}; [x + n] = [x] + n$$

**Lemma.2**

$$\forall x \in \mathbb{Z}; u(x) = u(b + x)$$

**Lemma.3**

$$\forall x, k \in \mathbb{Z}; f(x + bk) = ak + f(x)$$

**Lemma.4**

$$\forall x, k \in \mathbb{Z}; h(x + bk) = bk + h(x)$$